

CLL — Prognostic Markers, Cytogenetics & Staging



1. IGHV Mutated vs Unmutated

WHAT YOU SEE

Compass dial: MUTATED 60% indolent vs UNMUTATED 40% aggressive

WHAT IT ENCODES

IGHV MUTATED CLL (~60%) is INDOLENT with durable responses (good prognosis); UNMUTATED IGHV (~40%) is aggressive. Counterintuitive: 'mutated' = good here because it reflects post-germinal-center maturation.

BOARD TRAP

Backwards from most oncology: MUTATED IGHV is the GOOD prognosis. Unmutated IGHV predicts shorter time to treatment and worse outcomes.

2. FCR for Mutated IGHV

WHAT YOU SEE

'FCR DURABLE' near mutated/indolent side

WHAT IT ENCODES

Young/fit patients with IGHV-MUTATED and no TP53 abnormality can achieve very long remissions ('functional cure') with FCR (fludarabine, cyclophosphamide, rituximab). One of few chemo niches remaining.

BOARD TRAP

FCR is only appropriate for mutated-IGHV, TP53-wildtype, fit patients — giving it to unmutated/TP53-mutated patients yields poor durability and risks therapy-related MDS/AML.

3. del(13q) Sole — Best

WHAT YOU SEE

'del(13q) SOLE BEST' green/favorable region of map

WHAT IT ENCODES

Isolated del(13q) is the MOST FAVORABLE CLL cytogenetic abnormality — indolent course, near-normal life expectancy. Targets miR-15a/16-1. Most common single lesion.

BOARD TRAP

del(13q) is good ONLY when it is the sole abnormality — combined with other lesions the prognosis worsens. Don't equate 'has a deletion' with bad.

4. Trisomy 12 — Intermediate

WHAT YOU SEE

'TRISOMY 12' intermediate (yellow) zone of map

WHAT IT ENCODES

Trisomy 12 confers INTERMEDIATE prognosis. Associated with atypical morphology and NOTCH1 mutations. More common in atypical CLL.

BOARD TRAP

Trisomy 12 is intermediate — not lumped with the favorable del(13q) nor the adverse del(17p)/del(11q).

5. del(11q) — Adverse (ATM)

WHAT YOU SEE

'del(11q)' near volcano/adverse zone, ATM label

WHAT IT ENCODES

del(11q) targets the ATM gene → bulky lymphadenopathy and adverse prognosis. Historically chemo-resistant; BTK inhibitors largely overcome this poor risk.

BOARD TRAP

del(11q) = bulky nodes and aggressive disease, but BTK inhibitors (ibrutinib) neutralize much of its adverse impact — so it's less feared than del(17p)/TP53.

6. del(17p) / TP53 — Worst

WHAT YOU SEE

'del(17p)' red zone, '-70%', TP53 -4 PTS volcano

WHAT IT ENCODES

del(17p) and/or TP53 mutation is the WORST prognostic marker and predicts chemoimmunotherapy RESISTANCE. Mandates targeted therapy (BTK inhibitor or venetoclax) — never FCR. Test before every line.

BOARD TRAP

del(17p)/TP53-mutated CLL must NOT receive FCR/chemo — it's refractory. This is the single most important pre-treatment test; missing it leads to ineffective chemo.

7. CLL-IPI Scoring

WHAT YOU SEE

Right scoreboard: TP53 4pts, IGHV unmutated 2pts, β2M>3.5 2pts, Rai I-IV 1pt, Age>65 1pt

WHAT IT ENCODES

CLL-IPI integrates 5 factors: TP53 status (4), IGHV unmutated (2), β2-microglobulin >3.5 (2), clinical stage (1), age >65 (1). Stratifies low/intermediate/high/very-high risk with distinct survival.

BOARD TRAP

TP53 weighs most heavily (4 points) — it dominates the score. Don't treat all factors equally; TP53 alone can push a patient to high risk.

8. β2-Microglobulin >3.5

WHAT YOU SEE

'β2M >3.5 = 2 PTS' on scoreboard

WHAT IT ENCODES

Elevated β2-microglobulin (>3.5 mg/L) reflects tumor burden/turnover and is an independent adverse prognostic factor in CLL-IPI (2 points). Also prognostic in myeloma and lymphoma.

BOARD TRAP

β2M can be falsely elevated in renal impairment — interpret with caution in CKD where it accumulates independent of tumor burden.

9. Rai Staging (0–IV)

WHAT YOU SEE

Ladder/steps: 0 lymphocytosis, I nodes, II spleen, III anemia, IV low platelets

WHAT IT ENCODES

Rai stage 0 = lymphocytosis only; I = + lymphadenopathy; II = + organomegaly (spleen/liver); III = anemia <11; IV = low platelets. Stages III-IV (cytopenias from marrow failure) are high-risk.

BOARD TRAP

Stage III/IV requires cytopenia from MARROW INFILTRATION — autoimmune cytopenia does NOT upstage. Distinguish marrow failure from immune destruction.

10. Watch and Wait Standard

WHAT YOU SEE

'WATCH AND WAIT = STANDARD' top banner, order board

WHAT IT ENCODES

Asymptomatic early-stage CLL (Rai 0-II without active disease) is observed — early treatment does NOT prolong survival. Treat only for active disease (progressive cytopenias, bulky/symptomatic nodes, B symptoms, rapid lymphocyte doubling).

BOARD TRAP

Never treat CLL based on lymphocyte count or stage alone — a high WBC without symptoms is still watch-and-wait. Treatment indication is active/symptomatic disease.

11. Venetoclax in Adverse Disease

WHAT YOU SEE

'VENETOCLAX' label bridging favorable/adverse, miRNA region

WHAT IT ENCODES

Venetoclax (BCL2 inhibitor) is highly effective across risk groups including TP53-aberrant CLL, achieving deep MRD-negative remissions, often combined with obinutuzumab as fixed-duration therapy.

BOARD TRAP

Venetoclax requires aggressive tumor lysis syndrome prophylaxis with gradual ramp-up dosing — rapid escalation in high-burden CLL can be fatal.

12. NOTCH1 Mutation

WHAT YOU SEE

'NOTCH1' label near trisomy 12 / map

WHAT IT ENCODES

NOTCH1 mutations associate with trisomy 12, unmutated IGHV, Richter transformation risk, and resistance to anti-CD20 (rituximab) benefit. Adverse molecular marker.

BOARD TRAP

NOTCH1-mutated CLL derives less benefit from adding rituximab to chemo — a molecular caveat beyond the standard FISH panel.